

Muhammed Fayis K

Full-Stack Developer

fayiskooni@gmail.com | +91 9207892745 | [linkedin.com/in/muhammed-fayis-kooni](https://www.linkedin.com/in/muhammed-fayis-kooni) | github.com/fayiskooni | fayiskooni.xyz

Skills

Languages: JavaScript, TypeScript

Frontend: React, Next.js (App Router), Tailwind CSS, Shadcn UI, Radix UI, DaisyUI, HTML, CSS

Backend: Node.js, Express, REST APIs, WebSockets, Convex

Databases: PostgreSQL, MongoDB

Tools & Platforms: Git, GitHub, Vercel, JWT, Clerk, Stream SDK, Inngest, TanStack Query, Zustand

Concepts: SaaS Architecture, Multi-Tenant Systems, Authentication, RAG Pipelines, Real-Time Applications

Projects

Real-Time Language Exchange Platform

Live | [GitHub](#)

React 19, Node.js, Express, MongoDB, Stream Chat, Stream Video, JWT, TanStack Query, Zustand, DaisyUI

- Engineered a full-stack language exchange platform connecting users across 2 languages (native and target), with friend requests, real-time text chat, and video calls in a single interface
- Integrated Stream Chat and Stream Video SDKs with a custom Node.js backend that generates secure per-user tokens and syncs user profiles to Stream on signup and onboarding
- Secured authentication via HttpOnly JWT cookies to eliminate XSS-based token theft, with a protected route middleware verifying every incoming session request
- Architected server state using TanStack React Query for caching friend lists and user data, and developed 10+ reusable components including global theme switching via Zustand

Echo — AI Customer Support SaaS

Live | [GitHub](#)

Next.js 15, React 19, Convex, Gemini, Groq, Clerk, Stripe, Vapi, Shadcn UI, Turborepo

- Developed a multi-tenant B2B SaaS platform supporting 3 embed types (HTML, React, Next.js) via a lightweight JS snippet that injects a cross-origin iframe with postMessage communication for seamless widget integration
- Engineered a RAG pipeline using Gemini text-embedding-004 to generate 768-dimensional vectors from uploaded documents, retrieving the top 5 context chunks per query from Convex's native vector database
- Isolated tenant data by partitioning the vector store and all database queries by Clerk's verified orgId, preventing cross-tenant data leakage across organizations
- Implemented an event-driven billing system using Svix-verified webhooks to sync subscription state between Clerk and Convex, gating premium features behind active plan enforcement

Family Tree Generator

Live | [GitHub](#)

React 19, Node.js, Express, PostgreSQL, React Flow, ELK.js, TanStack Query, Zustand, DaisyUI, Shadcn UI

- Developed a full-stack genealogy platform where users manage 3 relationship types (members, marriages, parent-child links) visualized as a pannable and zoomable interactive tree
- Designed a 6-table normalized PostgreSQL schema with a dedicated parent_child table linking couple_id to child_id, correctly modeling half-siblings and multi-union lineages without recursive joins
- Transformed flat relational records into a Directed Acyclic Graph (DAG) using ELK.js layered layout algorithm with React Flow, supporting both vertical and horizontal collision-free tree rendering
- Enforced backend integrity checks preventing children from being assigned to unregistered couples or duplicate parent nodes, secured via HttpOnly JWT cookie authentication

Education

Bharathidasan University (BDU)

2022–2025

Bachelor of Arts (BA), English